# Naturality of homotopies: H(x) concatenated with ap of g along p equals ap of f along p concatenated with H(y)

The Univalent Foundations Program

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*Editorial note.* The counted statement is the author lemma labelled lem:htpy-natural (source lines 1026-1038), including the author's commutative diagram rendering of the assertion; the proof span 1039-1044 is the author's proof environment, retained line for line.

*Editorial note.* The ap shorthand used by the statement is the author's notation for the action of a function on a path, introduced immediately before the lemma; the compiled card resolves it through the book's own retained preamble macros.

*Editorial note.* The statement and proof cite no external labels, so no external references are declared.

*Editorial scope.* One original result of the HoTT book chapter cha:basics, counted once: naturality of homotopies with respect to the action of functions on paths. Only the counted lemma and its author proof are retained; the surrounding commentary and the displayed commutative diagram context remain uncounted except as part of the counted statement.

*Allowed standard prerequisites (editorial).*

- Homotopies: for functions f, g from A to B, an element H of the type product over x of f(x) = g(x), called a homotopy from f to g.
- The action of a function on paths: for f from A to B and p in x = y a path in f(x) = f(y), written ap of f applied to p, computing on refl by definition.
- Path concatenation with its definitional behaviour on refl.
- The induction principle for identity types (path induction) with its conversion rule.

**Lemma 0.0.1.** *Suppose $H : f \sim g$ is a homotopy between functions $f, g : A \to B$ and let $p : x =_A y$. Then we have*

$$H(x) \cdot g(p) = f(p) \cdot H(y).$$

*We may also draw this as a commutative diagram:*

$$\begin{array}{ccc} f(x) & \overset{f(p)}{=\!=} & f(y) \\ {\scriptstyle H(x)}\Big\| & & \Big\|{\scriptstyle H(y)} \\ g(x) & \underset{g(p)}{=\!=} & g(y) \end{array}$$

*Proof.* By induction, we may assume $p$ is $\mathsf{refl}_x$. Since $\mathsf{ap}_f$ and $\mathsf{ap}_g$ compute on reflexivity, in this case what we must show is

$$H(x) \cdot \mathsf{refl}_{g(x)} = \mathsf{refl}_{f(x)} \cdot H(x).$$

But this follows since both sides are equal to $H(x)$. □